

The iPSS Agent: Redefining Power System Simulation through Agentic AI

The Paradigm Shift: From Software Tools to Agentic Knowledge Work

The knowledge work landscape has crossed a critical inflection point known as the **"2025 December Shift"**. This transition marks the evolution of AI from a generative assistant into a system capable of **autonomous delegation**. We are moving away from human-centric models—where experts manually navigate fragmented software interfaces—toward an **agentic paradigm** where AI executes complex, multi-stage engineering workflows.

Defining Agentic AI In this architecture, AI functions as an "agent" capable of:

- **Autonomous Goal Pursuit:** Operating toward high-level objectives with minimal human steering.
- **Multi-Step Planning:** Deconstructing complex engineering problems into logical execution sequences.
- **Tool Integration:** Leveraging external simulation engines, APIs, and data repositories.
- **Adaptation & Iteration:** Dynamically adjusting strategies during exceptions and refining results through internal validation loops.

The Transformation of Knowledge Work This shift redefines the engineer's role from a "doer" of micro-tasks to an **orchestrator of macro-outcomes**. However, this presents a strategic challenge: the **"junior-level work pipeline"**. As entry-level analytical and documentation tasks are automated, organizations must rethink how to train the next generation of experts who historically learned the "physics" of the business through this foundational work.

The Power Grid Planning Challenge

Power grid planning is a high-stakes domain serving as the ideal crucible for agentic AI. As the grid decarbonizes and decentralizes, the complexity of planning studies has scaled beyond the limits of manual human processing.

Deconstructing the Agentic Grid Architecture To manage this complexity, a specialized **multi-agent system** is required:

- **Planning Agent:** Manages study dependencies and workflow timelines.
- **Data Agents:** Ingest and clean SCADA, weather, and interconnection queue data.
- **Simulation Agents:** Interface with simulation engines, such as the InterPSS engine, to run studies and troubleshoot excetions.
- **Optimization Agents:** Identify least-cost transmission upgrades or optimal storage placement.
- **Regulatory/Reporting Agents:** Translate technical results into NERC-style compliance documentation.

InterPSS: The Agent-Native Architecture

InterPSS (iPSS) is designed for the agentic era, built on the **Eclipse Modeling Framework (EMF)**. It recognizes that while software technologies are ephemeral, the physics of the power grid remains stable.

- **The "Death of the GUI":** iPSS treats the Graphical User Interface as an optional verification tool, allowing software to drive the simulation core directly through an **Agentic CLI** or desktop tools.
- **Skill MD Files:** In a departure from human-centric documentation, the iPSS Agent utilizes **Markdown-based instruction sets**. These allow AI agents to interact with the simulation engine with zero human intervention.
- **Enterprise Capabilities:** iPSS offers a lightweight **Java core with a Python API**, capable of nation-scale grid models and validated reliability on-par with legacy vendors.

Technical Implementation & Strategic Outcomes

The **ipss-agent** acts as the bridge between Large Language Models (LLMs) and the InterPSS core simulation engine. It facilitates **"Macro-Action Engineering,"** reducing complex sequences to executable agent skills.

Discovery and Inspection Workflows The agent enables natural language discovery within massive simulation result datasets:

- **Load Flow:** "Find the lowest voltage bus".
- **Contingency Analysis:** "Find the top N-1 loaded branches".
- **Reporting:** "Generate a NERC TPL-001-5 style report from the results directory".

Scaling Through "Simulation Skills" In this era, the primary bottleneck is no longer software capability, but **"skill refinement"**. Organizations can treat proprietary planning processes as a library of **Skill MD files**, effectively turning institutional knowledge into continuously improving automated instructions. This results in rapid scenario generation, reduced administrative overhead, and the ability to manage grid complexity that exceeds human-only processing.